Review of Egor Jakimov's "MATP32 Project"

Simon Gustafsson

6th of November 2026

The report presents the maximal inequality of Fefferman and Stein in a careful and rigorous way, and the proofs are generally easy to follow.

The duality between $\exp L$ and $L \log^+ L$ after Remark 5 is important in the proof of (iii) and would perhaps fit better as a separate proposition. Chebyshev's inequality gets its own proposition while this is only listed as a fact.

Lemma 6 assumes f positive but is later applied to arbitrary f_k. Since $f_k^* = |f_k|^*$, it would be enough to assume $f_k \geq 0$ at the start of the proof of Theorem 7, or simply to drop the positivity assumption in Lemma 6.

I see why the order of the proof of Theorem 7 is necessary, since the case $p = r$ is needed for (ii), which in turn gives the remaining cases of (i). It would be easier to follow this non-linear structure with clearer markers and boundaries between the cases. For example, "This shows that (i) holds" on page 6 refers only to the case $p = r$ while I first read it as a claim about all cases of (i). Additionally, numbering the equations could be helpful to the reader.

There is a strong connection to the course contents, however references to the course could perhaps be replaced by references to the course literature, which would make the report more self-contained. The introduction could also say a few words about the motivation for these results. Finally, a short abstract and suitable title could add to the overall impression.

Note, there are some small typos, such as "the precisely the ones" (p. 1), "a fact about Orlicz" missing "spaces" (p. 2), a missing period after "the following definition" (p. 2), "this follows φ being positive" missing "from" (p. 3) and "disjoint dyadic cube" instead of "cubes" (p. 3).